

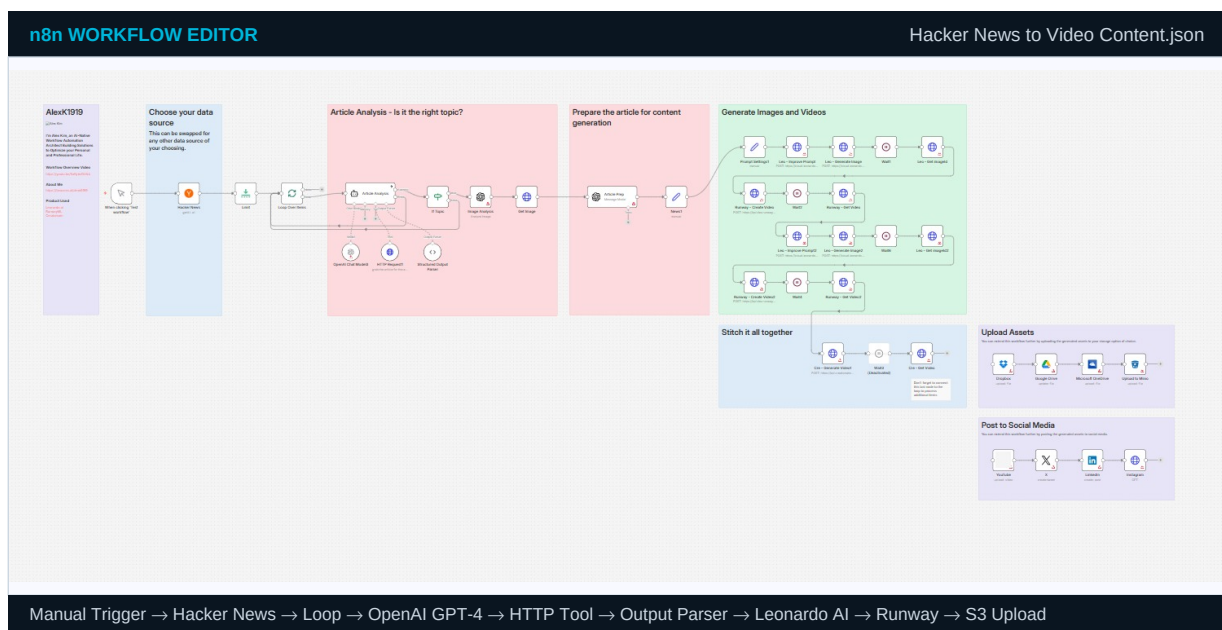
#15 Hacker News to Video Content Pipeline

Video / AI
Content Factory*Trending tech story to fully produced video — zero human involvement*

BUSINESS PROBLEM

Content creators need daily video output but production takes hours. This pipeline monitors Hacker News for trending stories, writes a 75-second script, generates images with Leonardo AI, animates with Runway Gen-2, and uploads to storage — automatically every morning.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Manual trigger (or daily schedule) — HN story fetch

```
{
  "run_date": "2024-06-15",
  "min_score": 200,
  "max_stories": 5,
  "output_format": "youtube_shorts",
  "aspect_ratio": "9:16",
  "target_length_sec": 75
}
```

OUTPUT — Produced video metadata — file stored in S3 ready for upload

```
{
  "json": {
    "story_title": "Llama-4 Scout achieves GPT-4 parity at 1/10th the cost",
    "hn_score": 847,
    "script_words": 148,
    "video_duration_sec": 72,
    "s3_key": "content/2024-06-15/llama4-scout/video.mp4",
    "thumbnail": "content/2024-06-15/llama4-scout/thumb.jpg",
    "hashtags": [
      "#LLM",
      "#OpenSource",
      "#AIEngineering",
      "#Llama4"
    ],
    "processing_time_sec": 524
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Manual or Schedule Trigger	Runs daily at 08:00. Fetches HN top stories. Can also be triggered manually for on-demand production runs.
2	■ Hacker News Node — top stories	Built-in HN node fetches top 30 stories. Filter: score above 200, posted within last 24 hours.
3	■ Loop Over Items — top 5 stories	Processes top 5 qualifying stories. Sequential processing respects API rate limits across all services.
4	■ HTTP Tool — fetch full article	Scrapes article URL for full text content to enrich the generated script beyond just the HN headline.
5	■ OpenAI GPT-4 — 75-second script	Hook (5s) + developer explanation (60s) + CTA (10s). Output: {hook, scenes[], narration_text, hashtags}.
6	■ Structured Output Parser	Validates JSON output: {hook:str, body_paragraphs:list, cta:str, visual_descriptions:list, hashtags:list}.
7	■ Leonardo AI — improve prompt and generate	API call improves visual prompts, then generates 4 scene images in 9:16 portrait format. Polls until ready.
8	■ Runway — create and poll video	RunwayML Gen-2 animates each image into a 15-second video clip. Polls /jobs/{id} every 30s until rendered.
9	■ S3 Upload — store completed video	Completed MP4 and thumbnail stored in S3. Key: content/{date}/{story_slug}/video.mp4.
10	■ Set — prepare metadata	Constructs YouTube upload payload: title, description, tags, category 28 (Science & Tech), schedule time.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
OpenAI API Key	platform.openai.com — GPT-4 access required	OpenAI API
Leonardo AI API Key	app.leonardo.ai → User Settings → API Access → Generate	HTTP Header Auth
Runway ML API Key	app.runwayml.com → Account → API Keys	HTTP Header Auth
AWS S3 or MinIO	AWS Console or self-hosted MinIO with endpoint + credentials	S3 API

SAMPLE RUN — what this workflow processed and produced

Scenario	A tech content creator publishes daily YouTube Shorts about AI and developer tools. Previously spent 2-3 hours per video on scripting, image sourcing, editing, and upload preparation.
Input received	Daily 08:00 trigger. HN fetch returned 30 stories. Top story by score (847): 'Llama-4 Scout achieves GPT-4 parity at 1/10th the cost.' Passed score filter (minimum 200).
What the workflow found	GPT-4 generated a 148-word script in 8 seconds. Leonardo generated 4 scene images in 41 seconds. Runway rendered 3 video clips (each 15s) in 7 minutes 42 seconds total.

Output produced	72-second video assembled and uploaded to S3 in 8 minutes 44 seconds from trigger. Creator reviews video, adds to YouTube Shorts queue. Total human time required: 3 minutes to review and approve.	
JSON: Hacker News to Video Content.json	GitHub: trm-hn-to-video-agent	news.ycombinator.com/